VERDANTIS AI: PLANT DISEASE CLASSIFICATION SYSTEM

6INTELSY

CS-301

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GROUP MEMBERS

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PROBLEM STATEMENT

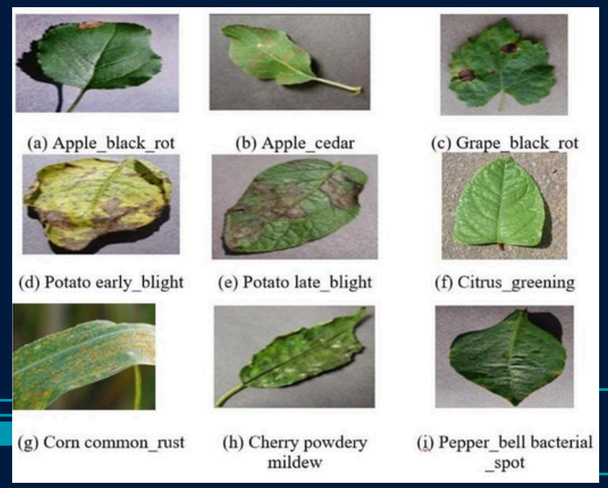
- Farmers and home plant owners struggle to identify plant diseases at an early stage
- Manual inspection is slow, subjective, and prone to errors
- There is a need for an automated, AI-based system to classify plant diseases from leaf images

OBJECTIVE & SYSTEM OVERVIEW

- Goal: Classify plant leaf diseases using images
- Input: Leaf image (captured via smartphone or uploaded)
- Output: Predicted disease class with confidence score
- System Pipeline:
  Image → CNN (MobileNetV2) → Prediction

DATASET & DATA PREPARATION

- Dataset: PlantVillage Dataset (38 classes, ~43,000 images)



- Data cleaning: Removed blurred and low-quality images

DATASET & DATA PREPARATION

- Data Split:
  - Train: PlantVillage dataset
  - Test: Mix of PlantVillage images and images sourced from Google Images

MODEL & TECHNOLOGY CHOICE

- Model: Convolutional Neural Network (MobileNetV2)
- Why this model is used:
  - Lightweight architecture suitable for mobile and low-resource devices
  - Fast training and inference, even on CPU-based systems
  - High performance in image classification tasks

MODEL & TECHNOLOGY CHOICE

- Techniques Used:
  - Transfer learning from a pretrained ImageNet model
  - Fine-tuning of upper layers to adapt to plant disease features

TRAINING APPROACH

- Framework: TensorFlow / Keras
- Training Strategy:
  - Step 1: Train only the classification head (base model frozen)
  - Step 2: Fine-tune last layers of the CNN for better feature adaptation

TRAINING APPROACH

- Data Handling:
  - Minimal data augmentation applied (e.g., basic flips and rotations)
  - Helps improve generalization without increasing model complexity significantly

RESULT & METRICS

Model Performance

- Achieved Accuracy: 0.87 – 0.96 across experiments
- Obtained Macro F1-score up to 0.95, ensuring balanced multi-class performance
- Strong precision (0.95–0.99) and recall (0.92–1.00) across 38 plant classes

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Macro F1-score: 0.9529

Confusion Matrix:
[[ 465    2    0 ...    2    0    0]
 [   3  475    0 ...    0    0    1]
 [   0    1  214 ...    0    0    0]
 ...
 [   0    0    0 ... 4263    6    0]
 [   0    0    0 ...    4  291    0]
 [   0    0    0 ...    0    1 1168]]

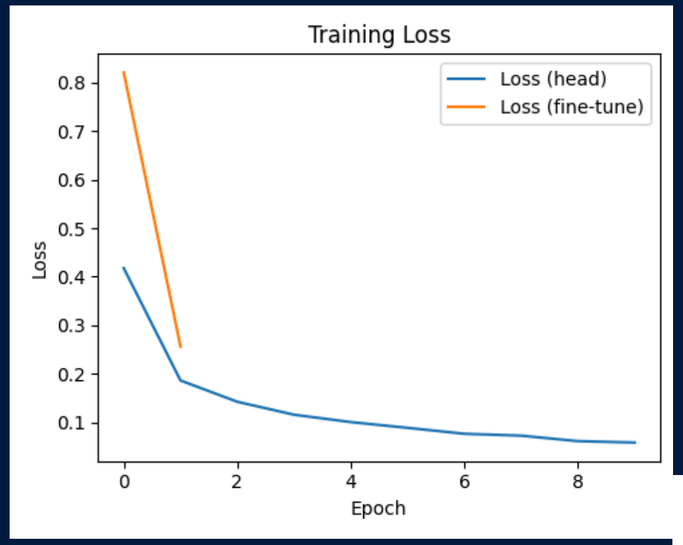
Classification Report:
                                                      precision    recall  f1-score   sup
port

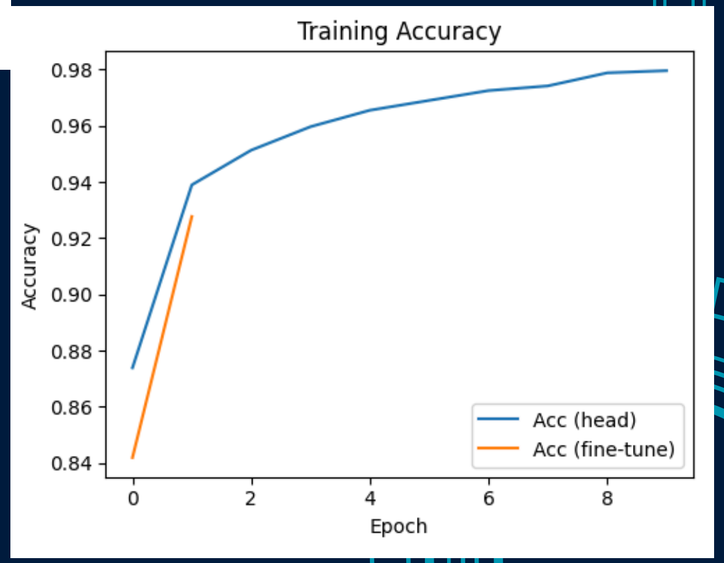
                                 Apple___Apple_scab       0.99      0.92      0.95
 504
                                  Apple___Black_rot       0.99      0.96      0.97
 496
                           Apple___Cedar_apple_rust       0.98      0.97      0.98
 220
                                    Apple___healthy       0.98      0.98      0.98
1316
                                Blueberry___healthy       0.97      1.00      0.98
1202
          Cherry_(including_sour)___Powdery_mildew       0.99      0.98      0.99
 842
                 Cherry_(including_sour)___healthy       0.99      0.98      0.98
 684
Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot       0.82      0.92      0.87
 410
                        Corn_(maize)___Common_rust_       0.99      0.99      0.99
 953
                 Corn_(maize)___Northern_Leaf_Blight       0.94      0.89      0.92
 788
                            Corn_(maize)___healthy       0.99      1.00      1.00
 929
                                  Grape___Black_rot       0.95      0.99      0.97
```

RESULT & METRICS

Training Behavior

- Model shows stable convergence with decreasing loss
- Fine-tuning improved feature extraction and overall accuracy
- Data augmentation enhanced generalization capability

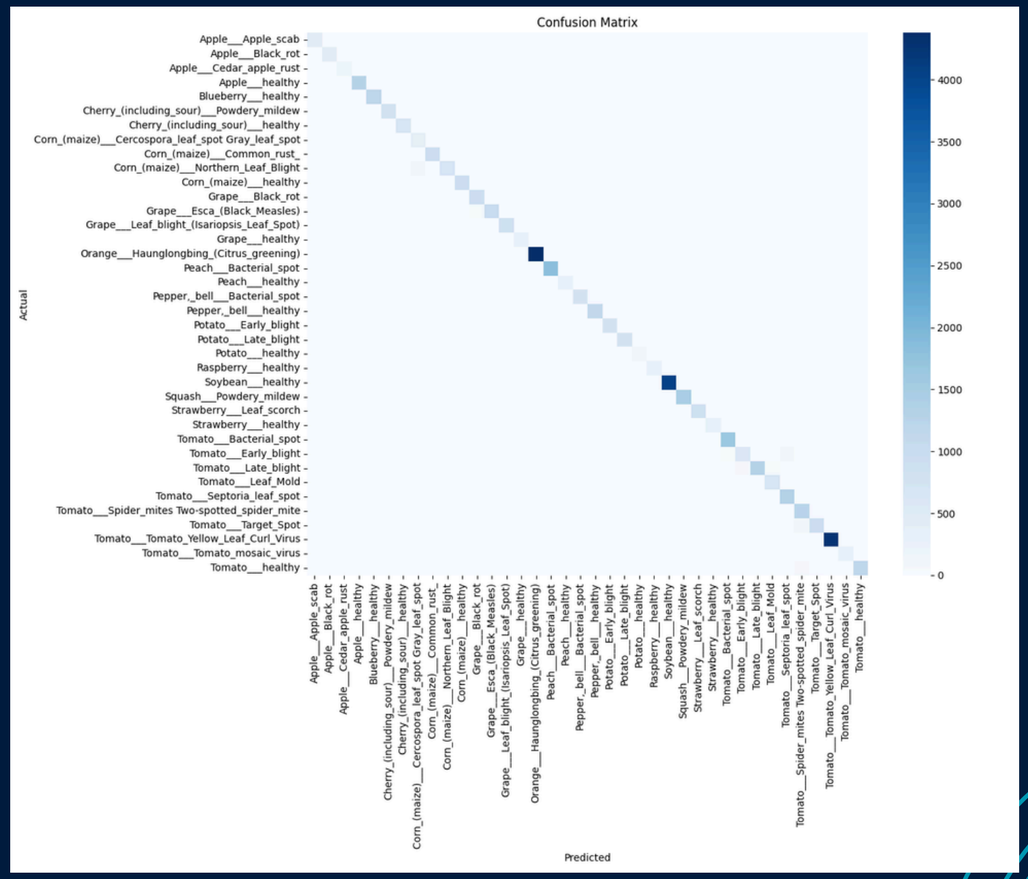




RESULT & METRICS

Evaluation Insights

- Confusion matrix shows high correct classification across most classes
- Majority of predictions fall along the diagonal (correct classifications)
- Indicates strong performance in distinguishing plant diseases



MODEL EVALUATION & ERROR ANALYSIS

Observations

- Model performs very well on clear and standard images
- Correctly identifies most plant diseases with high confidence
- Occasional errors occur in visually similar diseases

Limitations

- Sensitive to:
  - Lighting conditions
  - Image quality (blur, shadows)
  - Similar visual disease patterns

MODEL EVALUATION & ERROR ANALYSIS

Prediction: Strawberry___healthy (100.00%)

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Prediction: Pepper,_bell___Bacterial_spot (57.85%)

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Prediction: Strawberry___healthy (89.11%)

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Prediction: Cherry_(including_sour)___healthy (63.68%)

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MODEL EVALUATION & ERROR ANALYSIS

Intended Use & Expert Recommendation

- Suitable for home plant owners, and farmers for quick, preliminary disease checks.
- Not intended as a diagnostic tool for official research
- For accurate, research-grade identification, all results. must be verified by a plant expert.

DESIGN TRADE - OFFS & LIMITATIONS

Trade-offs (Unique points)

- MobileNetV2 chosen over heavier CNNs for faster inference and lower resource requirements.
- Slight compromise on peak accuracy is acceptable to prioritize real-time usability for farmers and home users.

Limitations (New content)

- PlantVillage dataset mostly controlled-environment images → limited real-world diversity.
- Performance can drop with unusual lighting, leaf angles, or background clutter.

SIMULATION

THANK YOU

"DETECTION FOR PLANT HOMEOWNERS, OR A COMPLEMENTARY TOOL FOR FARMERS — WHY CHOOSE ONE WHEN YOU CAN HAVE BOTH?" - TONY STARK

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